1

Real GDP vs. Nominal GDP

Gross Domestic Product (GDP) is the market value of all currently produced final goods and services in a given period of time (typically one year) by domestic and foreign supplied resources.

Nominal GDP (Y_n) is the value of currently produced **final goods** and services *measured* in **current year prices.**

Real GDP (Y_r) is the value of currently produced **final goods** and services *measured* in **constant prices** *(i.e. prices of the base year),* or nominal GDP adjusted for price level changes.

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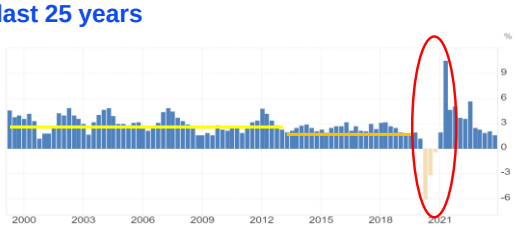
Australia's annual GDP Growth Rate for the last 25 years



Source: https://tradingeconomics.com/australia/gdp-growth-annual

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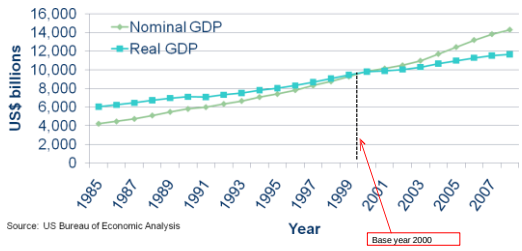
3

US Nominal GDP vs. Real GDP



Source: US Bureau of Economic Analysis

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4

Business cycle *[for the Short run (SR)]*

The periodic increases and decreased in overall economic activity are reflected in *production*, *employment*, *profits* and *prices*.

Normally **measured** with reference to **GDP.**

Composed of: **expansions** *(increases)* and **recessions** *(decreases)*

Or:

- *(recessions vs economic recovery)*
- *(economic upturns & downturns)*
- *(economic booms & bursts)*
- *(economic expansion & contraction)*

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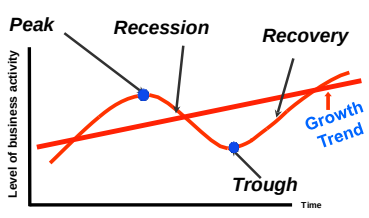
Phases of the business cycle



Sequence of business cycle: Peak -→ recession-→ trough-→ recovery →peak…
Phases: Recovery vs recession
Turning points: Peak vs trough

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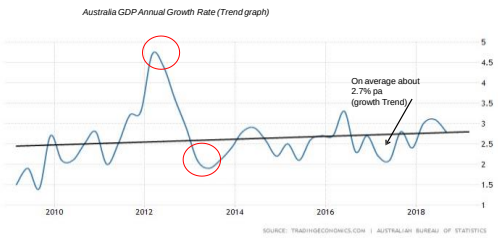
Australian GDP 2010-2019



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Business Cycle

GDP tends to increase over time but not in a linear fashion.

The economy exhibits cyclical growth which follows the business cycle. This has the following **phases** & turning pts:

1) A peak
2) **A recession**
3) A trough
4) **A recovery**

Australia's annual GDP Growth Rate 1961- Present

Source: https://tradingeconomics.com/australia/gdp-growth-annual

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Expenditure or Output Approach to Measuring GDP

Measuring overall **economic activity** by **adding** the **expenditure** on the **output produced** in the economy.

GDP = C + I + G + (X – M)

C=consumption
I=investment
G=government expenditure
X-M=net export (export-import)

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Personal Consumption Expenditures (C)

The total amount of **spending** by **consumers** on durable goods *(i.e. tangible)*, nondurable goods *(i.e.intangible goods)*, and services in a given period.

Key factors/indicators related to ***consumer spending (C):*** *consumer confidence, retail sales, new car registration, consumer credit, personal saving…)*

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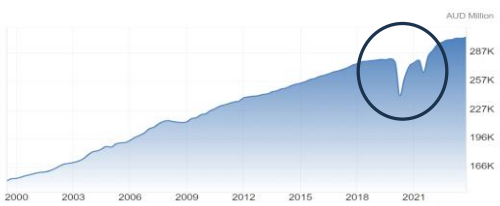
Australia Consumer Spending (C)



Source: https://tradingeconomics.com/australia/consumer-spending

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11

Gross Private Domestic Investment (I)

The total amount of spending on **nonresidential** structures, **equipment**, and **software**; **residential structures**; and **business inventories** in a given period.

(i.e. ***Capital stock*** *– machines, plant & equipment, tools etc… eg decrease in mining investment lately, residential & non-residential buildings)*

Key factors affecting ***Investment (I)*** *: Business confidence, Industrial production index, capacity utilisation,and production and Manufacturing index (PMI)*

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Australia fixed capital (I) formation

UTS

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Government Consumption Expenditures and Gross Investment (G)

The total amount of **spending** by *federal, state, and local* **governments** on consumption outlays for **goods and services** and for depreciation charges for existing structures and equipment and on **investment capital outlays** *(i.e. infrastructure)* for newly acquired structures and equipment in a given period.

(i.e. Fiscal Policy : Govt spending on health, education, age-care, child care, public capital investment (infrastructure), defence, public service… & taxation

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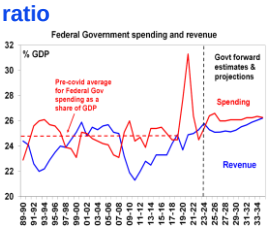
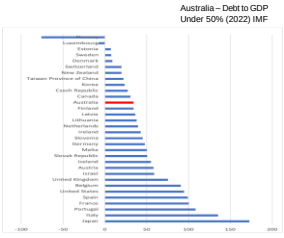
Australia Government spending (G) and Debt-to-GDP ratio



https://www.amp.com.au/insights-hub/blog/investing/economic-implications-fed-budget-2024-25



Debt to GDP ratio, advanced countries, %; Source: IMF (2022) World Economic Outlook Database, October 2022

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Net Export (NX) Spending (X – M)

The total amount of spending on **exports (X)** minus the total amount of spending on **imports (M)** in a given period.

(e.g., Exports of minerals (iron ore, coal), agricultural product, processed food, tourism, education while imports of consumer goods & capital goods).

Imports of travel, motor cars, manufactured goods

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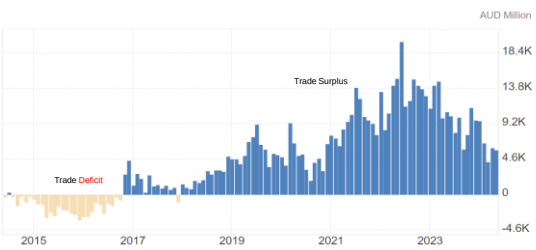
Australia net exports (NX) = exports (X) - imports (M)



https://tradingeconomics.com/australia/balance-of-trade

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The components of GDP (C, I, G, X, M)

GDP and its components in Australia in 2016.

GDP 2016	Total ($ billion)	Per capita ($)	Per cent of total
Consumption C	941	39,106	56.6 %
Investment I	417	17,205	25.1
Government G expenditures	304	12,654	18.3
Net exports (X-M)	-1	-43	-0.1
Total GDP	**1662**	**69,058**	100.0

https://econfix.wordpress.com/2014/08/17/aussie-still-relying-on-consumer-sector-2/

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Earnings or Income Approach to Measuring GDP

We measure overall economic activity by adding the **earnings** or **income** generated by selling the output produced in the economy.

Income approach :

GDP = *Compensation of employees* + *Proprietor's income* + *rental income* + *corporate profits* + net interest

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National Income (Y)

Compensation of employees: the ***wages and salaries*** and the fringe benefits paid by employers to employees.

Proprietors' income: the *income* of ***unincorporated businesses***, such as medical practices, law firms, small farms, and retail stores.

Rental income : the income households receive from the ***rental*** of their property (*or landlord i.e. owner of land*) .

Corporate profits: the *excess of* ***revenues*** *over* **costs** for the incorporated business sector of theeconomy. (*or business owner*)

Net interest: the interest private businesses pay to households for lending money to the firms minus the interest businesses receive plus interest earned from foreigners. (ie *owner of capital*)

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National Income (by the income approach)

COMPONENT	VALUE IN BILLIONS OF DOLLARS (% OF NATIONAL INCOME)
GROSS DOMESTIC PRODUCT	**13,807.5**
Less: Depreciation expenditures	1,618.0
Less: Statistical discrepancy (under)	+81.4
EQUALS: NATIONAL INCOME	**12,270.9**
Compensation of employees **(Wage)**	7,812.3 *(63.6%)*
Proprietor's income ***(Income fm svc)***	1,056.2 *(8.6%)*
Rental income ***(for land owners)***	40.0 *(0.3%)*
Corporate profits ***(TR – TC)***	1,642.4 *(13.4%)*
Net interest	664.4 *(5.4%)*

*National income = GDP - Depreciation expenditures + Statistical discrepancy

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Expenditure approach = Income approach

The calculation with <u>either</u> **approach** leads to the **<u>same</u> result;**

This is because they represent the **<u>two</u> sides of a transaction:** who **pays** *(a buyer, expenditure approach)* and who gets the **revenue** *(a seller, income approach).*

i.e. Total expenditure (TE) = total income (Y)

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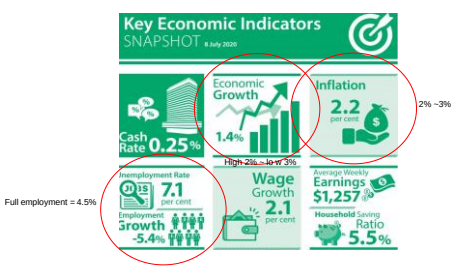
Economic indicators



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Other Macroeconomic Indicators: Price level and unemployment

Price level (PL) measures *(…inflation rates)*

- GDP deflator
- Consumer price index *(CPI)*
- Wholesale price index *(WPI)*
- Producer price index *(PPI)*

Employment and **Unemployment** *(unemployment rates)*

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GDP Deflator

The **GDP deflator** compares the **price** of **each year's output** of real goods and services to the **price** of that **same output** in a **base year**.

It is a broad **measure** of **price changes** because it reflects the overall changes in the prices of all of the components of GDP (C+I+G+X-M)

GDP deflator = $(Y_n / Y_r) \times 100$

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The GDP Deflator: Nominal GDP and Real GDP

VARIABLE	2010	2011
Nominal GDP *(Yn)*	$9,817.0 billion	$10,128.0 billion
Percent Change		3.17
Real GDP *( )*	$9,817.0 billion	$9,890.7 billion
Percent Change		0.76
GDP Deflator (price changes)	100	102.40
Percent Change (%)		2.4 %

***GDP Deflator = Yn/Yr x 100**

***GDP Deflator (2010 and 2011) = 10128/ 9890.7 * 100**

https://tradingeconomics.com/australia/gdp-deflator

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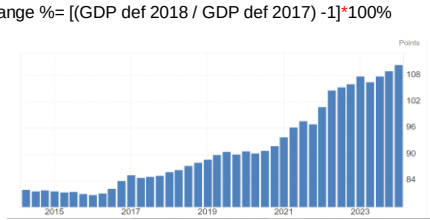
GDP Deflator

GDP deflator= (nominal GDP/real GDP)*100

Percentage change %= [(GDP def 2018 / GDP def 2017) -1]*100%

Australia GDP Deflator



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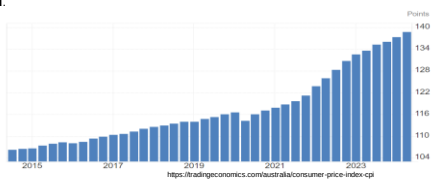
Consumer Price Index (CPI)

A measure of the combined **price** consumers pay for a **fixed market basket of goods and services** in **a given period** relative to the combined price of an identical basket of goods and services **in a base period**.

Australia's CPI:



https://tradingeconomics.com/australia/consumer-price-index-cpi

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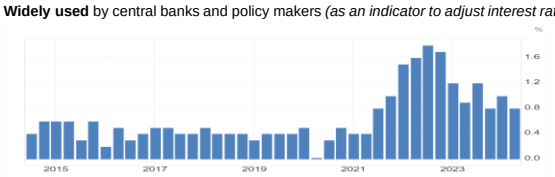
Core rate of inflation *(core inflation rate, trimmed mean inflation rate)*

Excludes **energy** and food **prices**; *(the most volatile items) that* are typically influenced by extraordinary or international factors *(e.g. weather, oil price).*

Widely used by central banks and policy makers *(as an indicator to adjust interest rates).*



https://tradingeconomics.com/australia/core-inflation-rate

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Notes on *"Inflation Target 2%~3%"*

The Governor and the Treasurer have agreed that the appropriate target for monetary policy in Australia is to achieve an inflation rate of 2–3 per cent, on average, over time.

*This approach allows a role for **monetary policy** in dampening the fluctuations in output (GDP) over the course of the (business) cycle. When **aggregate demand (AD)** in the economy is weak, for example, **inflationary pressures** are likely to be diminishing and **monetary policy** can be eased (expansionary), which will give a **short-term** stimulus to economic activity.*

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Unemployment: types

Frictional

- workers 'between jobs', temporarily laid off due to seasonality, and new entrants
- inevitable and partly desirable

Structural

- mismatch in skills and geographic location
- not employable without additional training, education and geographical movement.

Cyclical

- unemployment caused by the business cycle, or due to insufficient aggregate demand or total spending.

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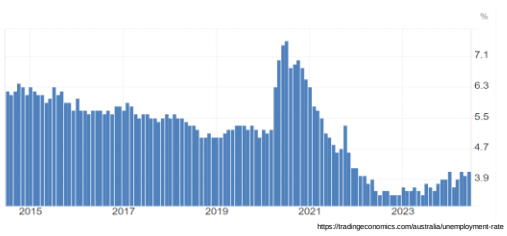
Australia Unemployment Rate (%)



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Measures of Employment and Unemployment

- Labour force (LF)
- Labour force participation rate %(LFPR%)
- Number employed (E)
- Number unemployed (UE)
- Unemployment rate % (UE%, ue%)
- Discouraged workers

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Labor Force (LF)

The civilian labor force is composed of those individuals **16 years of age** and **over** who are working in a job **(employed, E)** or who are actively seeking employment **(unemployed, UE)**

i.e. LF = E + UE

Labour force participation rate (LFPR%) = (LF ÷ adult population) x 100%

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Employed, Unemployed and unemployment rate

Employed (E): Persons **16 years of age** (15 in Australia) and over who, in the survey week, did **any work** as an **employee**, worked in their own business, profession, or farm; or worked without pay in a family business or farm.

Unemployed (UE): Persons **16 years** of age and over who **do not currently have a job**, but who are **actively seeking employment.**

Unemployment Rate (UR%): Proportion of the **labor force** that is **unemployed**.

*UR = (number of unemployed (UE) ÷ labor force (LF)) x 100%

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Discouraged Workers

Persons 16 years of age (15 in Australia) and over who are not currently **seeking work** because they believe that jobs in their area or line of work are unavailable or that they would not qualify for existing job openings.

Also defined as **marginally attached workers;**

Problem typically related to **long-term unemployment.**

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Australian labour force

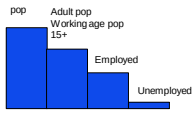
For June 24:

Employed persons ('000) =14406.05

Unemployed persons ('000) = 608.21

Unemployment rate (%) = 4.1%

Participation rate (%) = 66.90%



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Class Exercise: Unemployment in Australia (Sep 2016)

Population ('000)	23,940,000
Adult Population or Working Age Pop (WAP)	???
Employed (E)	11,947,000
Unemployed (UE)	???
Labour force (LF)	12,652,000
LFPR%	64.51%
UR%	???

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Class Exercise: Unemployment in Australia (Sept 2016)

Population ('000)	23,940,000
Adult Population or Working Age Pop (WAP)	**19,612,463**
Employed (E)	11,947,000
Unemployed (UE)	**705,000**
Labour force (LF)	12,652,000
LFPR%	64.51%
UR%	**5.57%**

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Natural Rate of Unemployment (Y_{nr})

Output at full employment (Y_f) = potential output (Y_p)= max output (about 5% UR)

The **minimum level of unemployment** that can be achieved with current institutions *without* ***causing inflation to accelerate***.

Also defined as **NAIRU**: Non-Accelerating Inflation Rate of Unemployment.

In the 90's it was estimated to be around 5-6% in the US.

Concept and estimations are somewhat controversial with economists divided on its usefulness and current value.

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